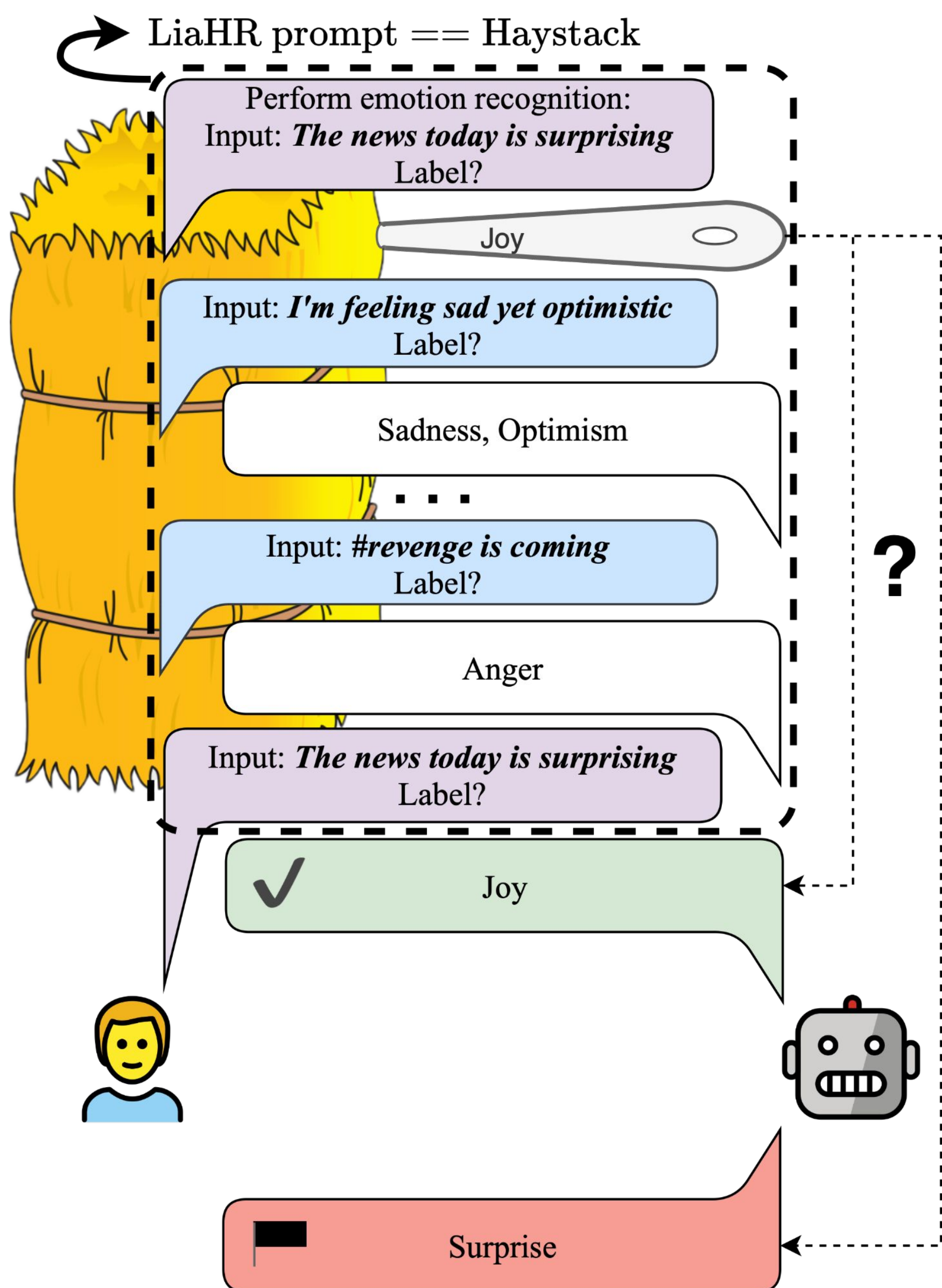


Subjective Label Verifiers

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TL;DR



Some problems do not have clear defined correct answers, making annotation quality assurance difficult at scale (cannot use verifiers, annotator or expert disagreement). To solve, we propose the Label in a Haystack prompt, where an example appears twice. In LLMs with strong zero-shot capabilities, Label in a Haystack creates a **tug of war** between: the first example and the prior of the model. If label is low-probability per the model, it overrides that label in the output, giving us a signal that the label may contain an error, and proposing an alternative. Otherwise, even if not the top choice of the model, the label persists. We also define proxy properties for subjective label verifiers, and evaluate their effectiveness in multiple datasets, including with human studies.

Datasets

SemEval 2018 Task 1 E-c: Multilabel emotion, 11 emotions
GoEmotions: Multilabel emotion, 7 emotions, annotator labels
MFRC: Multilabel morality, 6 moral dimensions, annotator labels
QueerReclaimLex: binary hate labels, in-group and out-group authors



Methodology (SIMPLE!)

- (1) Add the query and its candidate label – gold or random – as the first demo.
- (2) Add intermediate demos from the dataset
- (3) Add the query w/o label and predict → **LiaHR**. When the model predicts a different label than the candidate in the prompt, we consider it flagged. The model also proposes an alternative, which we evaluate for correctness.

Proxy Properties

Properties that can be evaluated **[semi]automatically** (without running verification tests, eg ecological validity, human studies)

- (1) **Nonconformity**: Flag some labels as unreasonable, but only a small percentage of the dataset
- (2) **Diversity**: The model should accept different labels for the same example consistently
- (3) **Noise Rejection**: High rejection rates for random labels
- (4) **Rectification**: When given random labels, the model should propose an alternative that is close to the original label

Results

Proxy Properties

Latest and bigger models meet them, **BUT**

- Too much RLHF and verifiable rewards degrades performance (eg GPT-3.5 better behavior than GPT-4)
- Rectification is not achieved by any model, only tendency for bigger models is observed
- Smaller models are too accepting of anything
- Different, disagreeing annotators have similar rejection rates, but need their regular examples are needed in between to learn annotator model

Human Studies

Humans flag some examples as reasonable and prefer LLM alternatives to original. But humans don't like regular inference of models!

Ecological Validity

BERT-based models trained on corrected labels show better or equal performance

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